

KONGJU NATIONAL UNIVERSITY

EKF vs UKF

Comparison in a Nonlinear Tracking System

Week 4 Assignment

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PROGRAM

M.S. in Mechanical Engineering

COURSE

Autonomous Driving

CODE REPOSITORY

https://github.com/Chal29/Autonomous/tree/main/Autonomous%20assignment

Q1. Baseline Comparison of EKF and UKF

I compared the EKF and UKF using the same simulated range-bearing measurements over 200 time steps. I repeated the experiment with random seeds 42, 43, and 44.

Table 1. Baseline performance across three runs.

Metric	EKF	UKF
Mean position RMSE (m)	0.24487	0.25137
RMSE standard deviation (m)	0.02216	0.01519
Mean NEES	3.17896	3.14501

The EKF had a slightly lower mean RMSE, by 0.00650 m, while the UKF showed less variation between runs. During the closest approach to the sensor (9-11 s), the RMSE was 0.15436 m for the EKF and 0.15532 m for the UKF. Both filters therefore performed similarly. Three runs do not establish a general advantage.

Q2. Effect of Removing Bearing-Angle Normalization

I repeated the experiment with angle wrapping disabled (--nowrap), using the same three seeds.

Table 2. Effect of disabling bearing wrapping.

Metric	EKF	UKF
RMSE with wrapping ON (m)	0.24487	0.25137
RMSE with wrapping OFF (m)	8.23637	8.22927
Mean NEES, wrapping OFF	1843.74683	1836.47423

The RMSE increased by approximately 33.6 times for the EKF and 32.7 times for the UKF. The first raw innovations exceeding π in magnitude occurred at steps 2, 1, and 6 for seeds 42, 43, and 44. For seed 42, an EKF innovation of −6.1283 radians at step 2 becomes +0.1549 radians after normalization.

Without wrapping, a small angular difference across the ±π boundary appears as a large error, causing an incorrect correction and severe inconsistency. Both filters were similarly affected.

Q3. Sensitivity to Initial Covariance

I scaled the initial state covariance by 1, 10, and 100 and tested each setting with seeds 42, 43, and 44.

Table 3. Initial covariance and estimation accuracy.

Scale	UKF α	EKF RMSE (m)	UKF RMSE (m)
×1	Default	0.24487	0.25137
×10	Default	0.25806	0.35437
×100	Default	Not reported	Failed at step 2
×100	1.0	0.26445	0.39444

Increasing the covariance from ×1 to ×10 raised the EKF RMSE by 5.4% and the UKF RMSE by 41.0%. The UKF was more sensitive.

At ×100 with the default parameters, the UKF failed at step 2 for all three seeds because its innovation covariance was not positive definite. This stopped the joint experiment. Repeating it with α = 1.0 allowed all runs to finish. This suggests that sigma-point scaling affected numerical stability under large uncertainty; this failure is implementation-specific.

Q4. Sensitivity to the UKF Scaling Parameter α

I tested five α values using the baseline covariance and the same three seeds.

Table 4. UKF performance for different α values.

α	Mean RMSE (m)	Mean NEES
0.001	0.25137	3.14501
0.01	0.25137	3.14501
0.1	0.25138	3.14493
0.5	0.25156	3.14382
1.0	0.25202	3.14188

All 15 runs completed. The RMSE range was only 0.00065 m (approximately 0.26%). Thus, α had little effect on baseline accuracy, but the ×100 experiment showed that it could affect numerical stability under high initial uncertainty.

Q5. NEES and Statistical Consistency

I used the Normalized Estimation Error Squared (NEES) to check whether each filter's estimated uncertainty agreed with its actual estimation error:

$$\mathrm{NEES}_k = (\hat{x}_k - x_k)^T P_k^{-1} (\hat{x}_k - x_k).$$

I averaged NEES at each time step over the three independent runs (seeds 42, 43, and 44). For the four-dimensional state, the expected NEES is 4. With three runs, the 95% pointwise chi-square interval for ANEES is approximately [1.468, 7.779].

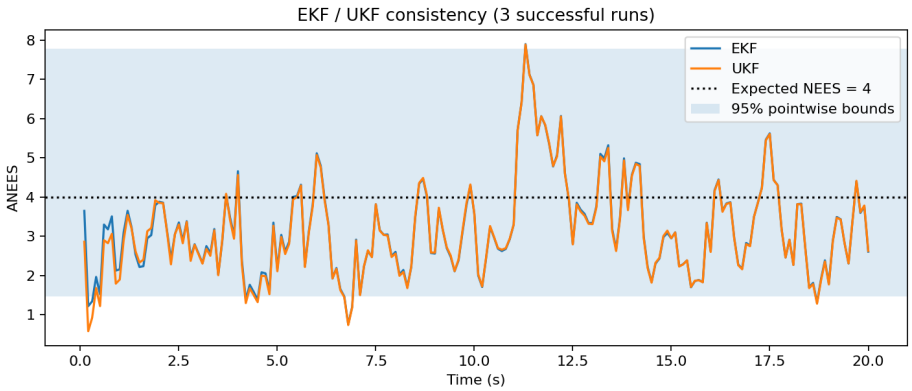


Figure 1. EKF and UKF ANEES over 200 time steps. The dotted line shows the expected NEES of 4; shading indicates the 95% pointwise bounds.

Table 5. Summary of consistency results.

Metric	EKF	UKF
Overall mean NEES	3.17896	3.14501
Steps within 95% bounds	191 / 200	190 / 200
Percentage within bounds	95.5%	95.0%

The ANEES curves almost overlap. Both briefly exceeded the upper bound at 11.3 seconds, while most other violations were below the lower bound. Their means were slightly below 4, suggesting mildly conservative covariance estimates.

Overall, both filters showed similar and reasonably consistent behavior in this scenario. However, only three independent runs were used, and consecutive time steps are temporally correlated. These results support consistency but do not prove it.

Q6. Information-Form Measurement Update and Commutativity Proof

I use $\Omega = \Sigma^{-1}$ (information matrix) and $\xi = \Omega\mu$ (information vector). Superscripts $-$ and $+$ denote prior and posterior values. R is measurement-noise covariance in this derivation (Q in the lecture notation).

1. Information matrix update using the Woodbury identity

For a linear observation $z = Hx + v$ with $v \sim N(0, R)$, the Kalman covariance update and gain are

$$\Sigma^{+} = (I - KH)\Sigma^{-},\ K = \Sigma^{-}H^{T}(H\Sigma^{-}H^{T} + R)^{-1}.$$

The Woodbury matrix identity is

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}.$$

Choose $A = \Omega^{-} = (\Sigma^{-})^{-1}$, $U = H^{T}$, $C = R^{-1}$, and $V = H$. Then

$$(\Omega^{-} + H^{T}R^{-1}H)^{-1} = \Sigma^{-} - \Sigma^{-}H^{T}(R + H\Sigma^{-}H^{T})^{-1}H\Sigma^{-}$$

$$= \Sigma^{-} - KH\Sigma^{-} = (I - KH)\Sigma^{-} = \Sigma^{+}.$$

Taking inverses gives the information matrix update:

$$\Omega^{+} = \Omega^{-} + H^{T}R^{-1}H.$$

2. Information vector update from the Gaussian posterior

The Gaussian prior has canonical form $p(x) \propto \exp[-\tfrac{1}{2}x^{T}\Omega^{-}x + x^{T}\xi^{-}]$. The likelihood is $p(z|x) \propto \exp[-\tfrac{1}{2}(z - Hx)^{T}R^{-1}(z - Hx)]$. Expanding and dropping terms independent of x gives

$$\log p(x|z) = -\tfrac{1}{2}x^{T}(\Omega^{-} + H^{T}R^{-1}H)x + x^{T}(\xi^{-} + H^{T}R^{-1}z) + C.$$

Matching the canonical coefficients yields $\xi^{+} = \xi^{-} + H^{T}R^{-1}z$. Recover $\Sigma^{+} = (\Omega^{+})^{-1}$ and $\mu^{+} = (\Omega^{+})^{-1}\xi^{+}$.

3. Nonlinear measurement: fixed linearization

For $z = h(x) + v$, linearize about the same predicted mean $\bar{\mu}$:

$$h(x) \approx h(\bar{\mu}) + H(x - \bar{\mu});\ \tilde{z} = z - h(\bar{\mu}) + H\bar{\mu}.$$

The update is $\Omega^{+} = \Omega^{-} + H^{T}R^{-1}H$ and $\xi^{+} = \xi^{-} + H^{T}R^{-1}\tilde{z}$. The affine correction in $\tilde{z}$ must not be omitted.

4. Commutativity for two independent measurements

For conditionally independent measurements of the same state with fixed H_i and R_i, define $\Delta\Omega_i = H_i^{T}R_i^{-1}H_i$ and $\Delta\xi_i = H_i^{T}R_i^{-1}z_i$ (or use corrected $\tilde{z}_i$ for fixed linearizations).

Order 1 → 2: $\Omega_{12} = \Omega^{-} + \Delta\Omega_1 + \Delta\Omega_2$; $\xi_{12} = \xi^{-} + \Delta\xi_1 + \Delta\xi_2$.

Order 2 → 1: $\Omega_{21} = \Omega^{-} + \Delta\Omega_2 + \Delta\Omega_1$; $\xi_{21} = \xi^{-} + \Delta\xi_2 + \Delta\xi_1$.

Because addition is commutative, $\Omega_{12} = \Omega_{21}$ and $\xi_{12} = \xi_{21}$, so both orders give identical posterior covariance and mean. For a sequential EKF that relinearizes after each measurement, the intermediate estimates and Jacobians can differ by order, so the approximate posterior may also differ.

Conclusion. Information-form updates add precision and information-vector contributions. Their order is immaterial for independent linear-Gaussian measurements or nonlinear measurements with fixed linearization points.